

SWAPNIL SUNIL MALI

Software Development Engineer | Backend & Distributed Systems | Cloud (AWS / Azure) | AI & GenAI

Deerfield Beach, FL | +1 5615909087 | swapnilmali7971@gmail.com | linkedin.com/in/swapnil-mali-a1244919a

PROFESSIONAL SUMMARY

Software Development Engineer with 2 years at PricewaterhouseCoopers (PwC) designing, building, and operating production-grade backend services, REST APIs, and microservices on AWS and Azure. Owns features end-to-end, from low-level design, API contracts, and data modeling through containerized deployment, CI/CD, monitoring, and production incident response. Hands-on with AI and GenAI delivery, including a Retrieval-Augmented Generation (RAG) pipeline powering internal tooling and an end-to-end MLOps pipeline on Azure. Strong fundamentals in Python, JavaScript, SQL/NoSQL, data structures and algorithms, system design, and Agile/DevOps practice. M.S. Computer Science, Florida Atlantic University (Dec 2025).

TECHNICAL SKILLS

Development: Python, JavaScript, SQL, REST API Design, Microservices & Distributed Systems, System Design (HLD/LLD), Data Structures & Algorithms, Git

Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, IAM, API Gateway, CloudWatch, VPC), Microsoft Azure (DevOps, Pipelines, ML Studio), Docker, Kubernetes (EKS), CI/CD (Jenkins, Azure DevOps, GitHub Actions), Linux & Bash, Monitoring & Observability

AI / GenAI / ML: Generative AI & LLM Integration, Retrieval-Augmented Generation (RAG), Python ML Stack (Pandas, TensorFlow), MLOps (MLflow, Azure ML Studio)

Databases: PostgreSQL, MySQL, MongoDB (NoSQL), SQL/NoSQL Data Modeling

Engineering Practices: Agile/Scrum, Code & Design Reviews, Unit & Integration Testing, DevSecOps, Helm, ArgoCD, GOSU (Guidewire), Technical Documentation

WORK EXPERIENCE

Associate 2 | Software Development Engineer

Jul 2022 – Jun 2023

PricewaterhouseCoopers (PwC), Remote

- Designed and owned production-grade backend services and REST APIs end-to-end on AWS (EC2, Lambda, S3, RDS, IAM, API Gateway, VPC), covering low-level design, API contracts, and data modeling through to deployment, CloudWatch dashboards and alarms, and root-cause analysis of production incidents; supported 10+ application workloads and partnered with product managers and architects to cut requirement-to-deployment cycle time by 30% through Agile sprint planning and iterative design reviews.
- Led CI/CD and containerization, building pipeline-as-code templates in Jenkins and Azure DevOps for Docker and Kubernetes (EKS) deployments and embedding DevSecOps quality gates, automated security scanning, and secrets management across environments, increasing deployment frequency 3x, reducing release cycle time by 50% and post-release defects by 25%, and raising operational efficiency 60% across 5+ application teams.
- Built and integrated a Retrieval-Augmented Generation (RAG) pipeline connecting LLMs to existing backend services and workflows to power AI-enabled internal tooling, and authored Python automation for infrastructure orchestration, health checks, and cost reporting that was adopted as standard platform tooling across the DevOps organization.

Associate 1 | Software Developer

Jun 2021 – Jun 2022

PricewaterhouseCoopers (PwC), Remote

- Developed reusable, modular backend modules and automation services in GOSU (Guidewire) and JavaScript, driving peer code reviews and shared-component design, improving test execution performance by 50%, increasing automated code coverage by 35%, reducing duplicate code by ~40%, and cutting new-engineer onboarding time by 2 weeks.
- Operated and monitored AWS infrastructure (EC2, S3, IAM, RDS) for 10+ production applications, sustaining 99.8% uptime through proactive alerting, log analysis, and root-cause troubleshooting, while reducing cloud costs by 15% through architectural optimizations.
- Integrated continuous testing and deployment pipelines in Azure DevOps across quarterly release cycles, and built Python and Bash tooling for infrastructure provisioning, log analysis, and configuration management, saving 8+ engineering hours per week across the team.

ACADEMIC PROJECTS

Restaurant Sales Forecasting | ML System

Feb 2025 – Apr 2025

FAU Cloud Computing Capstone · Azure ML Studio

- Built and benchmarked ensemble forecasting models (XGBoost, LSTM, Prophet, SARIMAX, Random Forest, Elastic Net) across 100K+ restaurant order records, engineering features from Python and Pandas EDA that surfaced seasonality, outliers, and midday demand patterns, improving MAE by 18% and MAPE by 12% over baseline.
- Engineered an end-to-end MLOps pipeline on Azure with automated training, MLflow model versioning and experiment tracking, and CI/CD deployment through Azure DevOps and Azure ML Studio, authoring reusable Python modules for preprocessing, feature engineering, and pipeline orchestration that were adopted as project-wide templates; used GitHub Copilot, Claude Code, and ChatGPT to accelerate development, debugging, and documentation.

Cloud-Enabled AI Wearable Transcription App

Sep 2024 – Nov 2024

FAU Distributed Systems · iOS / CloudKit

- Designed and built a real-time speech transcription application for Apple Watch integrating Whisper AI, with low-latency distributed device-to-cloud communication through CloudKit and an API Gateway layer.
- Applied distributed systems principles and formal software architecture artifacts (use-case and sequence diagrams) to sustain production-grade responsiveness under constrained compute, memory, and network conditions.

EDUCATION

M.S. Computer Science

Jan 2024 – Dec 2025

Florida Atlantic University, Boca Raton, FL | GPA: 3.5 / 4.0

Coursework: Cloud Computing, Distributed Systems, Machine Learning, Deep Learning, AI, Analysis of Algorithms, Software Engineering

B.Tech, Electronics & Communication Engineering

Jul 2017 – May 2021

CMR Institute of Technology, Hyderabad, India | CGPA: 7.5 / 10

CERTIFICATIONS & CONTINUOUS LEARNING

- AWS Solutions Architect Associate (In Progress, Expected 2026)
- Actively upskilling: Platform Engineering · MLOps & AI/ML Deployment · Observability & OpenTelemetry · Kubernetes & Service Mesh · Go Programming · Distributed Systems Design